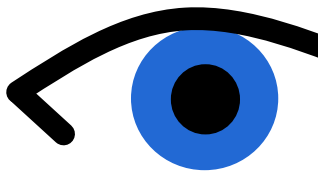
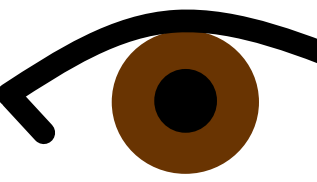
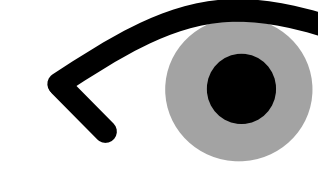


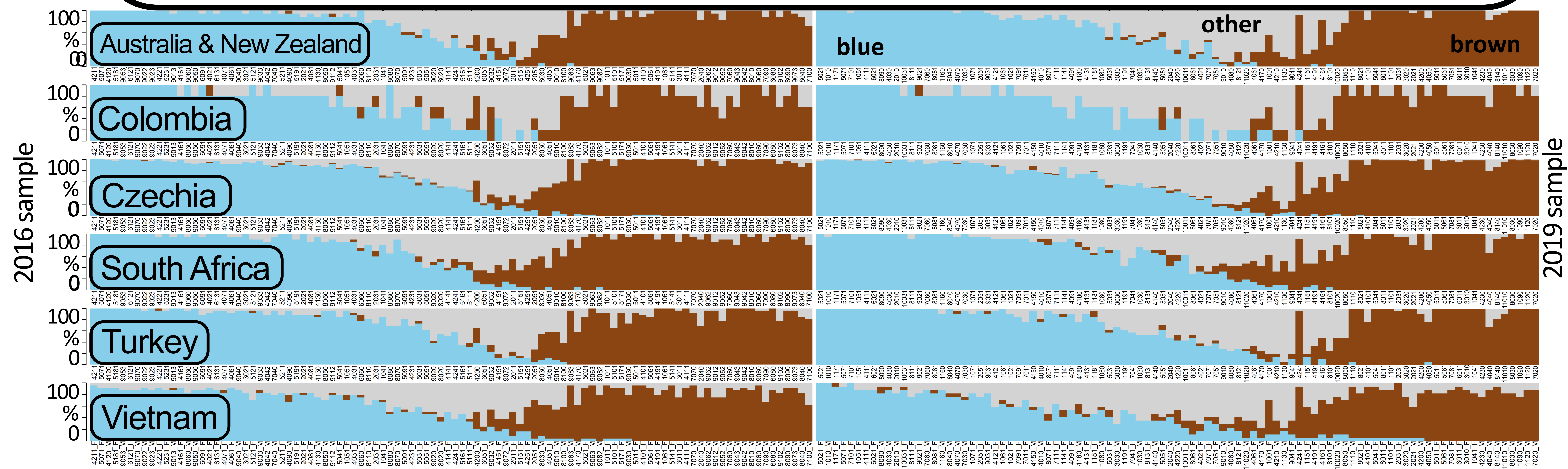
Cross-Cultural Universality of Eye-Colour Categories ?

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"Colour perception is not universal as it varies both between individuals (Bosten, 2022) and across cultures (Gibson et al., 2017)."
However; does it apply for eye (iris) colour?

We asked (in local language): What colour is this person's eyes (iris)  blue  brown  other

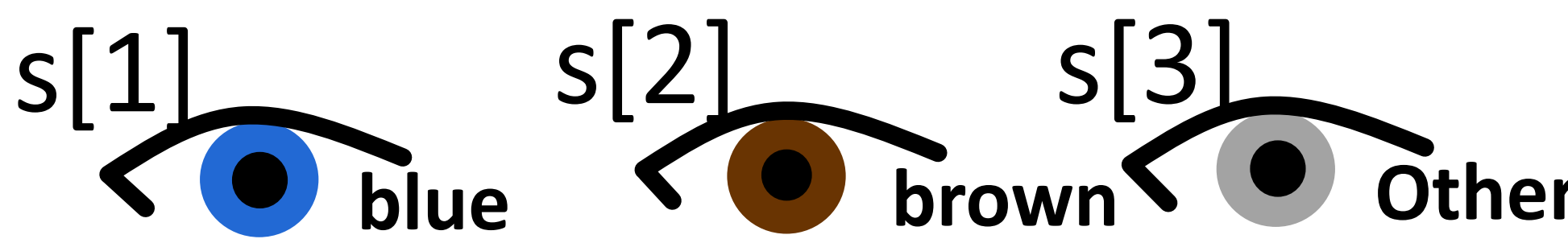
Each vertical bar is one face. The more of blue, brown, and "other" colour, the more it was rated as such in...



Participants in every country rated the same sample of 195 Czech stimuli. And they used the categories similarly.

Main Analysis: Softmax Regression

probability of rating the eye as:



lightness(measured L^*) +
redness(measured a^*) +
yellowness(measured b^*) +
Interactions(L^*a^*b)

When iris $L^* = +1$ SD (38.1); $a^* = -1$ SD (0.9); $b^* = -1$ SD (0.2) , which is pretty blue for an iris, the model predicts "Blue" being by far the most likely choice in every sample

Australia & New Zealand

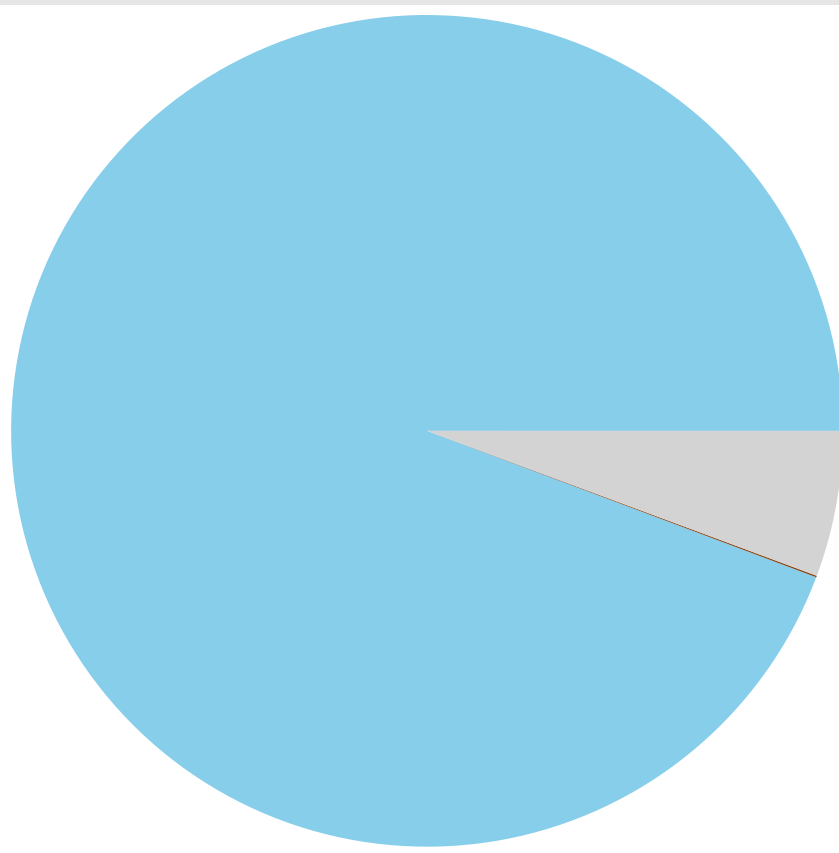
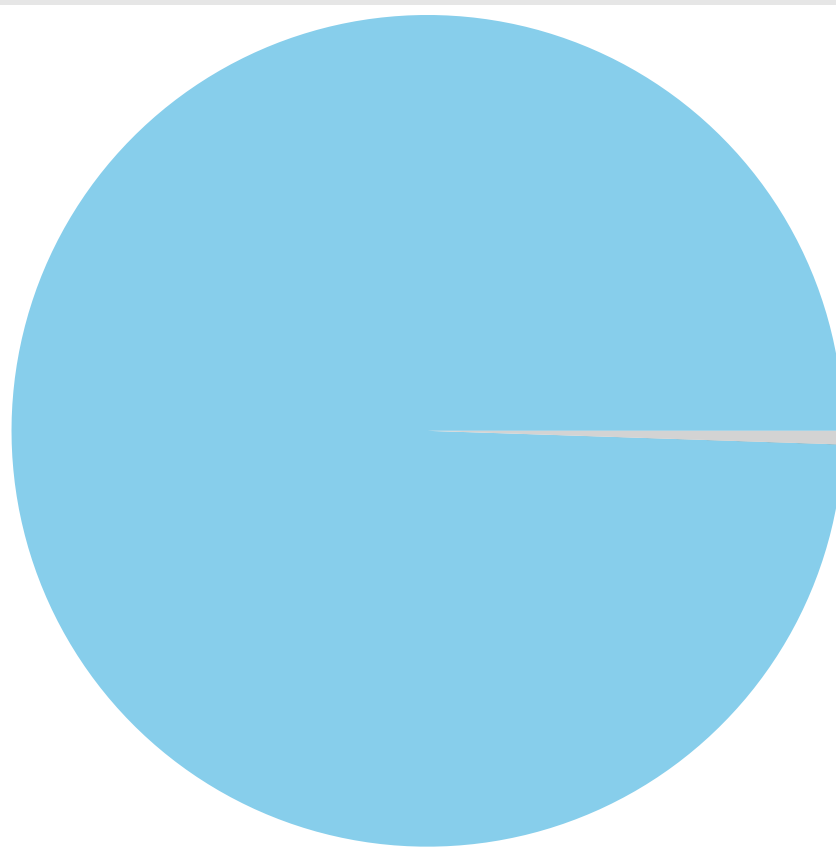
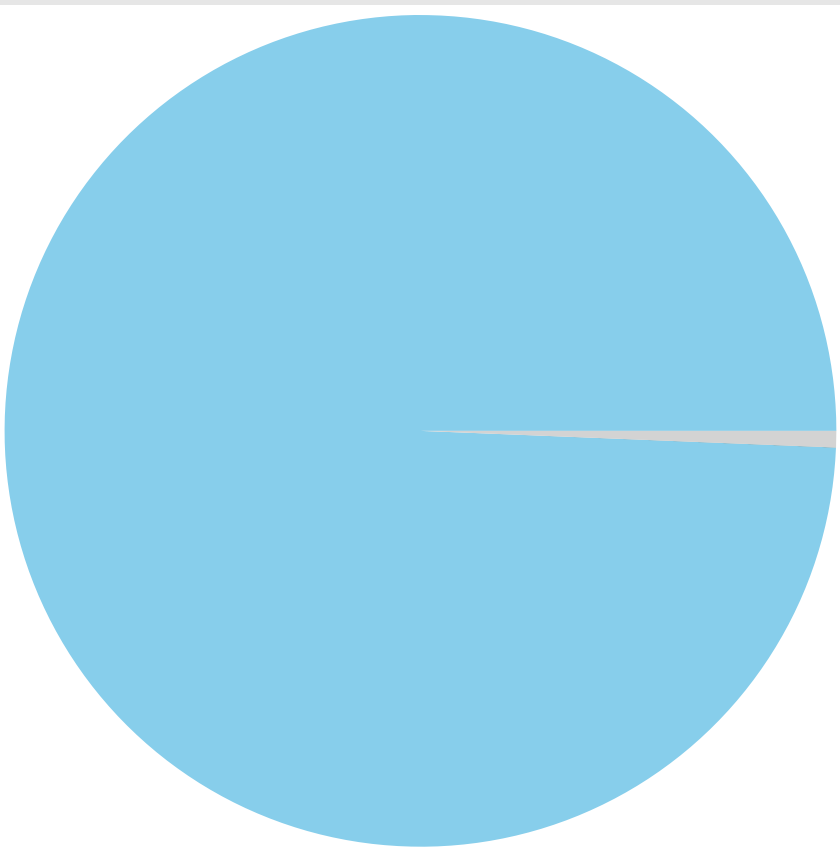
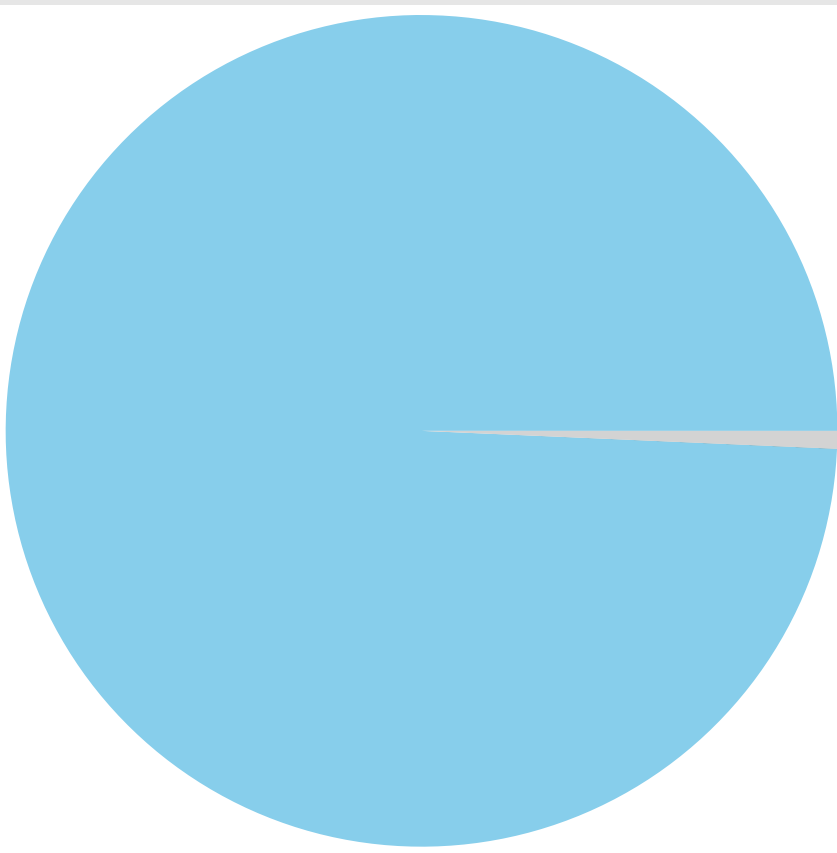
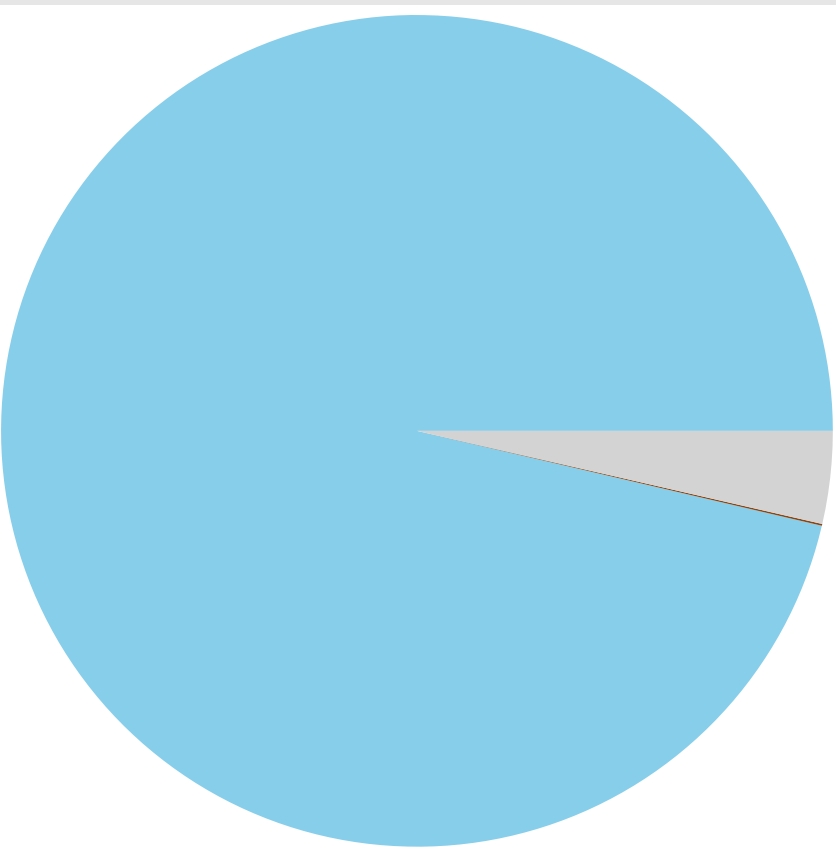
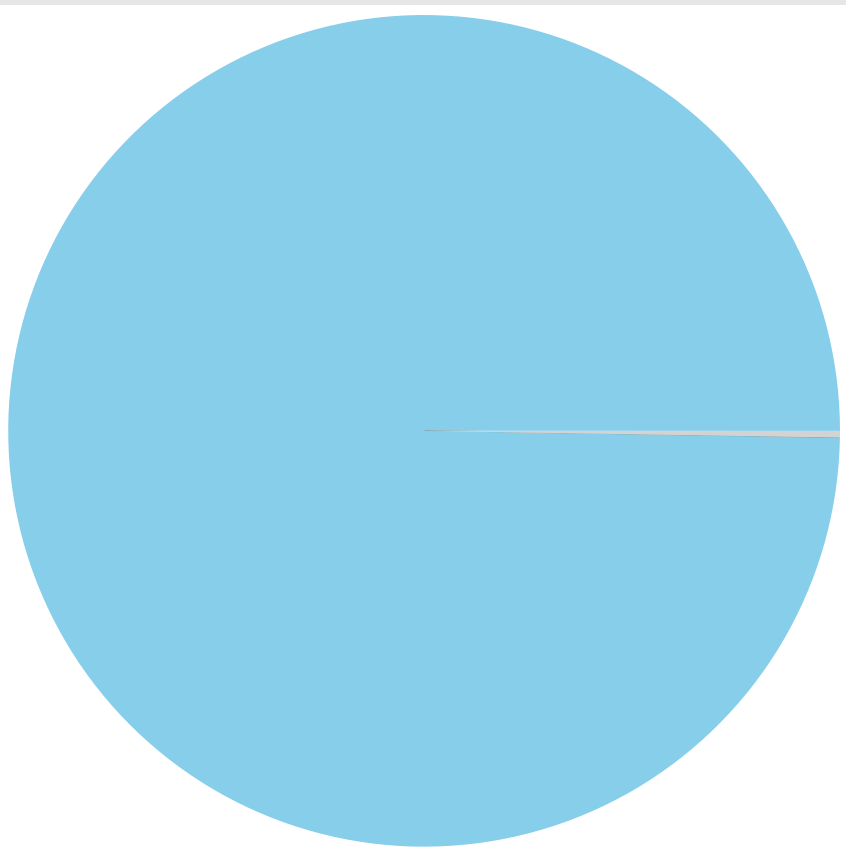
Colombia

Czechia

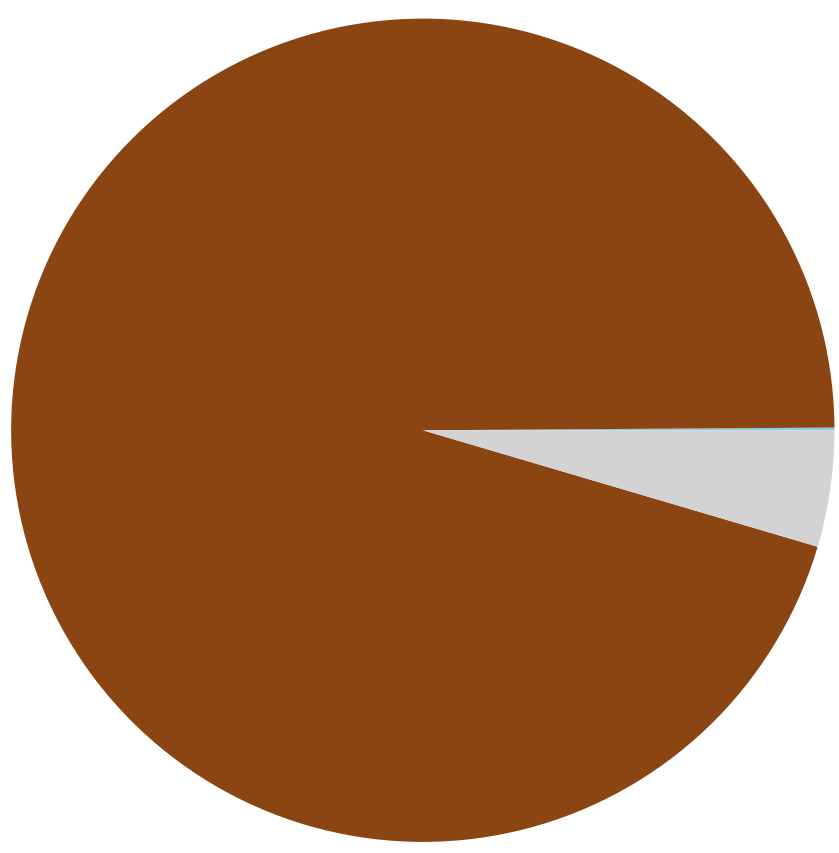
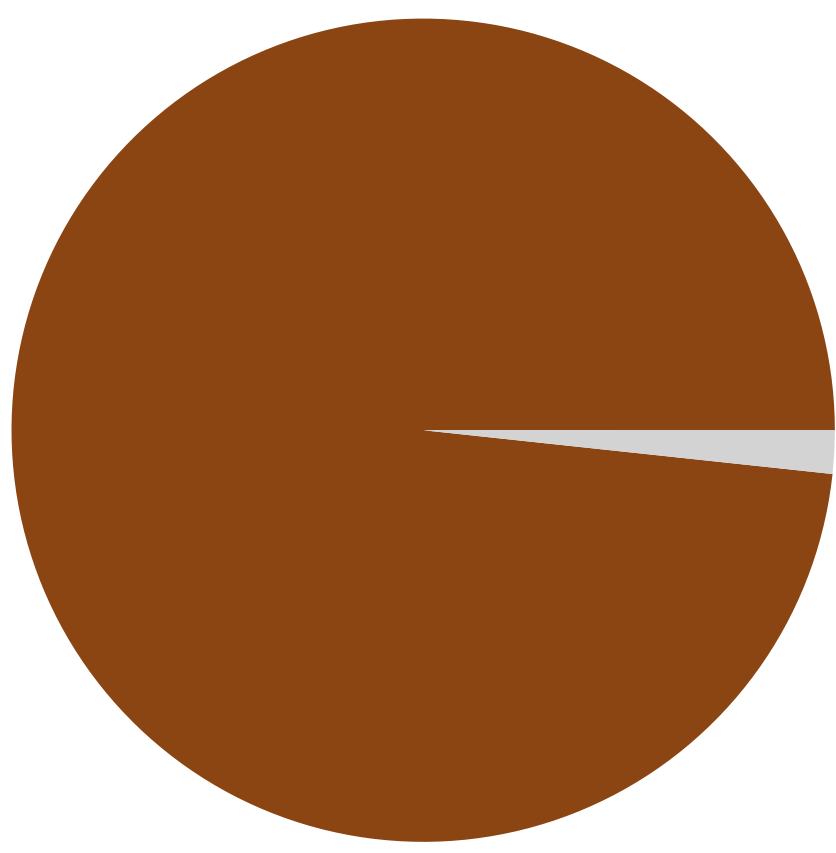
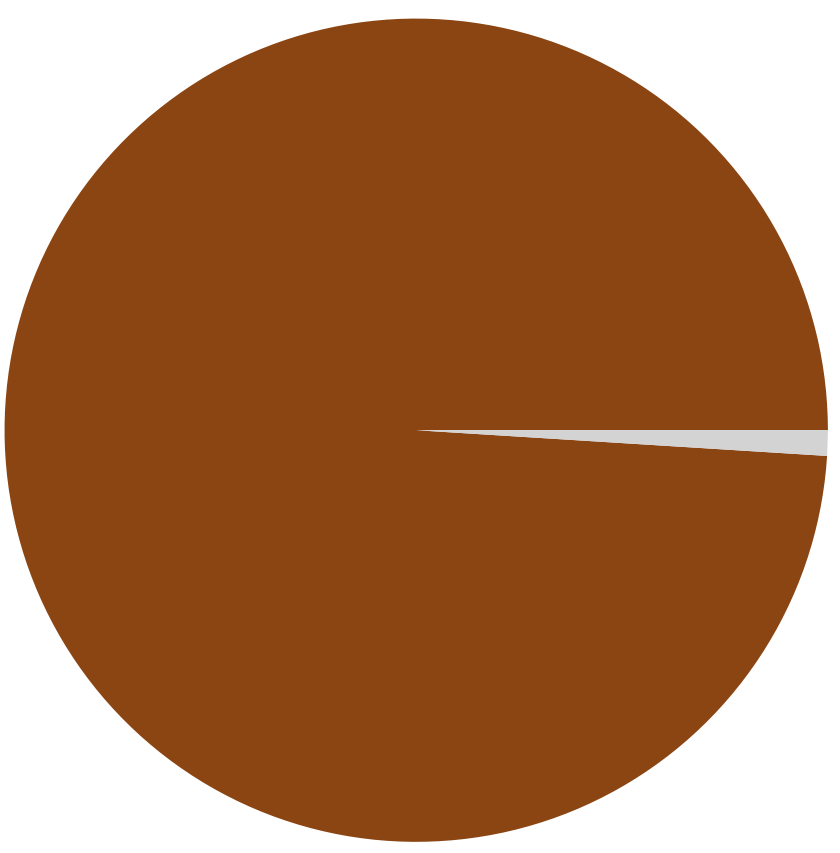
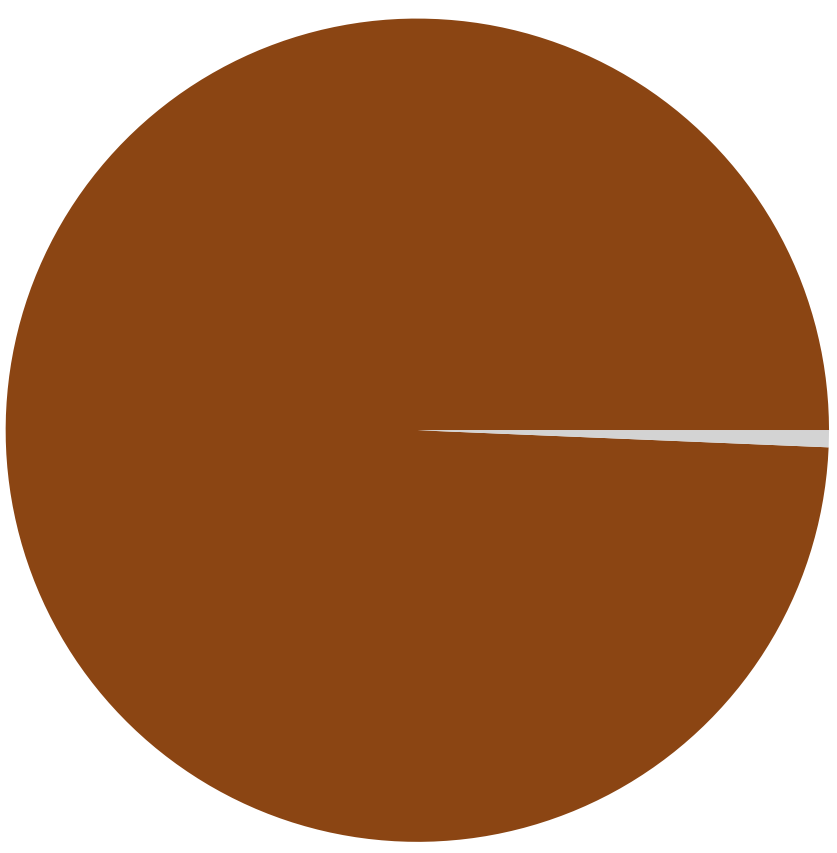
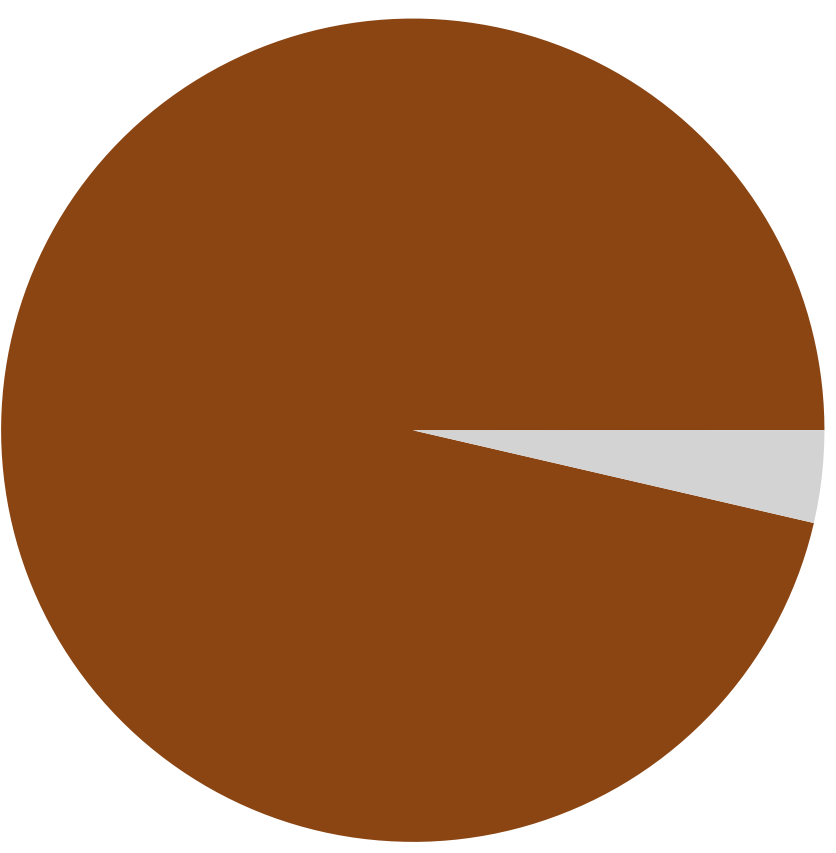
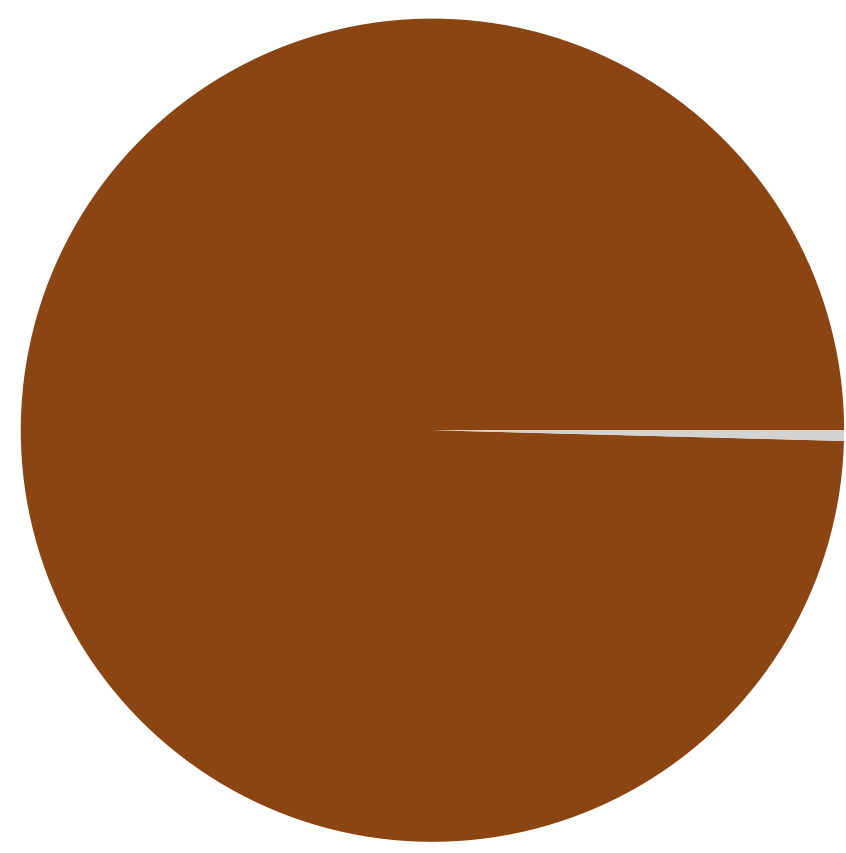
South Africa

Turkey

Vietnam



And when $L^* = -1$ SD (24.4); $a^* = 1$ SD (8.8); $b^* = 1$ SD (14) , which is pretty brown for an iris, the model predicts "Brown" being by far the most likely choice in every sample



Cross-Cultural Universality of Eye-Colour Categories ✓

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